THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 2: **Acids, Bases and Salts** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If H^+ concentration of a solution is 10^{-3} M, its pH is: 1

(a) 3 **(b)** 11 **(c)** -3 **(d)** 7

2. A solid X reacts with dilute HCl giving a gas that turns lime water milky; on strong heating, X gives Na_2CO_3. X is: 1

(a) NaCl **(b)** NaOH **(c)** Na_2SO_4 **(d)** $NaHCO_3$

3. Assertion (A): Plaster of Paris should be stored in moisture-proof containers. Reason (R): It sets into a hard mass of gypsum on contact with moisture. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is false but R is true.
(d) A is true but R is false.

4. Gypsum of molar mass 172 g loses 27 g on conversion to plaster of Paris. The mass of plaster of Paris obtained from 172 g is: 1

(a) 118 g **(b)** 126 g **(c)** 145 g **(d)** 172 g

5. A gas turns blue litmus red and is formed in the chlor-alkali process. The gas is: 1

(a) chlorine **(b)** oxygen **(c)** carbon dioxide **(d)** hydrogen

6. Assertion (A): Anhydrous $CuSO_4$ is used as a test for water. Reason (R): Anhydrous $CuSO_4$ is blue in colour. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. The percentage of water in $CuSO_4 \cdot 5H_2O$ (Cu = 63.5, S = 32, O = 16, H = 1) is approximately: 1

(a) 50% **(b)** 10% **(c)** 36% **(d)** 18%

8. A student finds pH of solutions P, Q, R, S as 2, 6, 8 and 13. Which two should be mixed to give the closest to neutral? 1

(a) P and S **(b)** Q and R **(c)** R and S **(d)** P and Q

9. Assertion (A): Tooth enamel is corroded when pH in the mouth falls below 5.5. Reason (R): Calcium phosphate is the hardest substance in the body and is insoluble in water. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. The mass of water of crystallisation in 286 g of washing soda is: 1

(a) 18 g **(b)** 90 g **(c)** 180 g **(d)** 106 g

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does the pH of rain fall below 5.6 in industrial areas? 2

12. If 5 mL of NaOH needs 20 mL of a dilute acid, how much acid is needed for 25 mL of NaOH? 2

13. Why is the pH of an aqueous solution of NH_4Cl less than 7? 2

14. Explain why a solution of baking soda is neither strongly acidic nor strongly basic. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why is baking soda better than washing soda for a cake? Give two reasons with the effect of heat. 3

16. Pure water does not conduct electricity but adding a little acid makes it conducting. Explain. 3

17. Name a compound used to make 3
(i) glass
(ii) a cast for fractured bones and write their formulae.

OR

Calculate the percentage of water of crystallisation in gypsum (Ca = 40, S = 32, O = 16, H = 1). 3

18. A reaction mixture of $NaOH$ and HCl becomes warm. Explain, with an equation and the nature of reaction. 3

OR

Explain how a decrease in pH of a river water affects aquatic life. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Calculate the mass of plaster of Paris formed from 860 g gypsum (Ca = 40, S = 32, O = 16, H = 1) and the mass of water lost. 5

OR

A test tube contains zinc and dilute HCl. A second contains Zn and NaOH. Compare observations, equations and explain the amphoteric nature of zinc. 5

20. Explain, with equations, the conversion NaCl -> NaHCO3 -> Na_2CO_3 -> $Na_2CO_3 \cdot 10\,H_2O$. Give a use for each. 5

OR

Explain, with reasoning, the colour of litmus, phenolphthalein and methyl orange in solutions of HCl, NaOH and NaCl. 5

END OF QUESTION PAPER — Total: 40 Marks